

KhanLab-BayArea

Web Accessibility



Avery Davis

Why does accessibility matter?



Inclusivity

Web Accessibility ensures that the internet can be used by all people regardless of disability or limitations. Everyone deserves to be able to access the internet equally.

Legal Compliance

Laws such as the Americans with Disabilities Act ensure that disabled members of society are not discriminated against. It is important that we comply with accessibility regulations.

Better UX for All

Accessibility features, such as good color contrast, clear labels, and keyboard navigation, improve usability for all users. Proper web accessibility makes websites easier to navigate for people on mobile devices, with slow internet speeds, etc.

Disability Categories



Visual

Blindness, low vision, color-blindness.



Auditory

Deafness and hard-of-hearing.



Motor

Inability to use a mouse, slow response time,
limited fine motor control.



Cognitive

Learning disabilities, distractibility, inability
to remember or focus on large amounts of
information.

Terminology

Ally

A11y is a shortened version of 'accessibility.' The 11 refers to the 11 letters in between A and y.

Ally Tree

A simplified version of the DOM (document object model) that provides semantic info for accessibility.

AT

Assistive Technology, including software and hardware that improve interaction with the web.

Reading Order

The logical sequence in which content is presented to assistive technologies.

WCAG

Web Content Accessibility Guidelines, the international standard for accessibility with three levels (A, AA, AAA).

Semantic HTML

The practice of using HTML elements according to their intended meaning instead of for visual appearance.

WCAG

WEB CONTENT ACCESSIBILITY GUIDELINES



WCAG: pour

Perceivable:

- Provide text alternatives for non-text content.
- Provide captions and other alternatives for multimedia.
- Create content that can be presented in different ways, including by assistive technologies, without losing meaning.
- Make it easier for users to see and hear content.

Operable:

- Make all functionality available from a keyboard.
- Give users enough time to read and use content.
- Do not use content that causes seizures or physical reactions.
- Help users navigate and find content.
- Make it easier to use inputs other than keyboard.

Understandable:

- Make text readable and understandable.
- Make content appear and operate in predictable ways.
- Help users avoid and correct mistakes.

Robust:

- Maximize compatibility with current and future user tools.

Key Takeaway:

Most of A11y is just good webdev practices!

Focuses of A11y

HTML

- Semantic HTML
- Landmarks
- Alternative Text

CSS

- Color & Contrast
- Focus Indicators
- Flexible Design

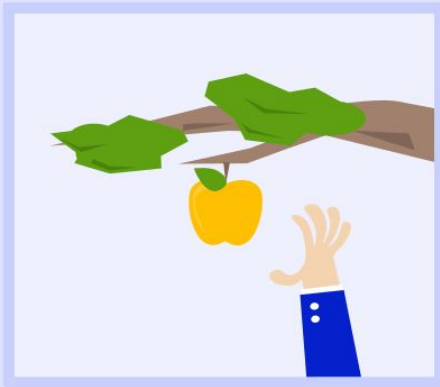
Other

- Frameworks (Bootstrap)
- JavaScript
- Templating (Jinja)

HTML

HTML: Alternative Text

IMAGE ALT TEXT FOR SEO



alt="yellow apple"



alt="hand reaching for
a yellow apple hanging
from the tree"

- Do not say “image of,” “photo of,” etc.
- End with a period
- Provide information that you lose without the image

CSS

CSS: Color & Contrast

Contrast Checker

[Home](#) > [Resources](#) > Contrast Checker

Foreground
Hex Value
#0000FF
Color Picker
Alpha
1
Lightness

Background
Hex Value
#FFFFFF
Color Picker
Lightness

Contrast Ratio
8.59:1
[permalink](#)

Normal Text

WCAG AA: **Pass**

WCAG AAA: **Pass**

The five boxing wizards jump quickly.

Large Text

WCAG AA: **Pass**

WCAG AAA: **Pass**

The five boxing wizards jump quickly.

Graphical Objects and User Interface Components

WCAG AA: **Pass**

★
Text Input

Contrast Checker

[Home](#) > [Resources](#) > Contrast Checker

Foreground
Hex Value
#FF3DDB
Color Picker
Alpha
1
Lightness

Background
Hex Value
#FFFFFF
Color Picker
Lightness

Contrast Ratio
3.02:1
[permalink](#)

Normal Text

WCAG AA: **Fail**

WCAG AAA: **Fail**

The five boxing wizards jump quickly.

Large Text

WCAG AA: **Pass**

WCAG AAA: **Fail**

The five boxing wizards jump quickly.

Graphical Objects and User Interface Components

WCAG AA: **Pass**

★
Text Input

CSS: Focus Indicators & Flexible Design

- Keyboard focus is essential
 - Example: <https://webaim.org/resources/contrastchecker/>
- Use rem (or em) for units
 - Element size will respond to user choices
 - Zoom in: text will be larger and still legible

Other

Bootstrap

<https://getbootstrap.com/docs/5.3/getting-started/accessibility/>

- It is fully possible to reach AAA compliance with Bootstrap
- Common a11y problems:
 - Color contrast
 - Visually hidden content
 - Motion

JavaScript & Jinja

JavaScript:

- Use it only when necessary
- Make sure it does not mess with the DOM or read order

Jinja:

- Make sure it renders properly

ARIAS

What are ARIAs?

- Accessible Rich Internet Applications
 - Supplement HTML to assist technologies
- Better to use semantic HTML over ARIAs
 - Only use when native HTML doesn't solve a problem
- No ARIA is better than a bad ARIA

ARIA basics



WCAG Checklist



Testing A11y

Auditing

Example:

<https://www.khanacademy.org/>

Key Steps:

1. Keyboard tab through entire site
2. Screen reader
3. Shrink page size
 - a. Zoom in/out



Automatic Testing

2 Main Options:

1. [W3 Checker](#)
2. [WAVE Evaluator \(extension\)](#)

Additional Resources

WCAG

<https://www.wcag.com/>

Allycasts

<https://www.youtube.com/playlist?list=PLNYkxOF6rcICWx0C9LVWWVqvHIYJyqw7g>

MDN: Accessibility on the Web

https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Accessibility

WebAIM checklist

<https://webaim.org/standards/wcag/checklist>

WCAG (Azul coding) checklist

<https://webaim.org/standards/wcag/checklist>